Al-Balqa' Applied University
Faculty of Engineering Technology - Department of Basic Scientific Sciences
Mid Exam - Linear Algebra

Name: الإجابات النموذجية Date 11 /8/2022

Part A: Multiple choice (12 marks)

-Be sure that you have 6 different questions.
-Complete the following table by putting (the correct answer letter) in the table.

Q1	Q2	Q3	Q4	Q5	Q6
b	a	c	b	a	d

Q1) Suppose that the augmented matrix for a system of linear equations has been reduced by row operation to the given form $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 2 \end{bmatrix}$ The solution of the linear system (respectively x,y,z) is

a) -4,3,-2 b) 1, 0, 1 c) -1,0,-1 d) 4, -3, 2

Q2) Determine the value of a for which the system $\begin{aligned} x+y+7z &= -7 \\ 2x+3y+17z &= 11 \\ x+2y+(a^2+1)z &= 6a \end{aligned}$ has no solution

a) -3 b) -2 c) 2 d) 3

Q3: If $(2I+4A)^{-1} = \begin{bmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{10} \end{bmatrix}$ *then* $tr(A) =$ a)4 b)14 c)3 d)6

Q4: Let the matrix A= $\begin{bmatrix} 1 & 2 & 0 \\ 3 & 5 & 4 \\ -1 & 2 & 6 \end{bmatrix}$ the cofactor C_{31} is a) -3 b) 8 c) 2 d) 6

Q5: If $A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 4$ *evaluate the* $\begin{vmatrix} 3(2a+g) & 3(2b+h) & 3(2c+i) \\ d & e & f \\ a & b & c \end{vmatrix} =$ a) -12 b) -1 c)9 d) 8

Q6: Let $A = \begin{bmatrix} 1 & 3 & 12 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{bmatrix}$ *Then* $|2A^{-1}| =$ a)3 b)1 c) 4 d) 2

Part B: Fill the blank (12 marks)

Q1: The values of the unknown constant in order for $A=\begin{bmatrix}2 & a-b & 5\\ 3 & 4 & 6\\ a & 6 & 3\end{bmatrix}$ to be symmetric is ---- $a=5$, $b=2$

Q2: *Let* $A^{-1}=\begin{bmatrix}2 & -5 & -1\\ 3 & 1 & 2\\ 5 & 0 & 4\end{bmatrix}$ *Then* $\left(3A^{T}\right)^{-1}=$ _____ $\begin{bmatrix}\frac{2}{3} & 1 & 5/3\\ -\frac{5}{3} & \frac{1}{3} & 0\\ -\frac{1}{3} & \frac{2}{3} & 4/3\end{bmatrix}$

Q3: The values of X in order for $A=\begin{bmatrix}x-4 & 4 & 6\\ 0 & x+3 & 0\\ 0 & 0 & x-5\end{bmatrix}$ to be invertible is $\mathbb{R}/\{4,-3,5\}$

Q4: The 3×3 matrix $A=[a_{ij}]$ whose entries satisfies the stated condition $a_{ij}=j-i$, A= $\begin{bmatrix}0 & 1 & 2\\ -1 & 0 & 1\\ -2 & -1 & 0\end{bmatrix}$

Q5: Let $A=\begin{bmatrix}1 & 5 & 6 & 2\\ 2 & 4 & -2 & 1\\ 6 & 12 & -6 & 3\\ 1 & 0 & -1 & 2\end{bmatrix}$ show that det (A)=0, without directly evaluating the determinant, by state the property which use.---------- $R_3=3R_2$

Q6: The values of k for which det(A)=0, where $A=\begin{bmatrix}k-4 & 0 & 0\\ 5 & k & 0\\ 3 & 1 & k+2\end{bmatrix}$ is---------- $k=4,0,-2$

Part C: Written questions (8 marks)

Question 1: For some nonsingular square matrix A, assume that $A^{-1}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ -2 & -1 & 2\end{bmatrix}$ solve the system $AX=\begin{bmatrix}1\\ 0\\ -1\end{bmatrix}$

Question 2: Find the augmented matrix for the given system of linear equation.

$$x-3y+z=4$$
$$y=-9z+7$$
$$x+3y=1$$

Question 3: Show that the matrix $\begin{bmatrix}1 & 0 & 0\\ 0 & \cos\theta & -\sin\theta\\ 0 & \sin\theta & \cos\theta\end{bmatrix}$ is invertible for all values of θ and find A^{-1} using adjoint.

With my best wishes __

عودة (-)

[1] $X = A^{-1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix}$$ (2)

[2] $$\begin{bmatrix} 1 & -3 & 1 & 4 \\ 0 & 1 & 9 & 7 \\ 1 & 3 & 0 & 1 \end{bmatrix}$$ (1)

[3] $A^{-1} = \dfrac{\text{adj } A}{|A|}$ (1)

(1)

$|A| = \cos^2\theta - -\sin^2\theta = 1$ so Invertible

$$\text{adj } A = [c_{ij}]^T = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}^T$$ (2)

$$A^{-1} = \frac{\text{adj } A}{1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}^T$$ (1)